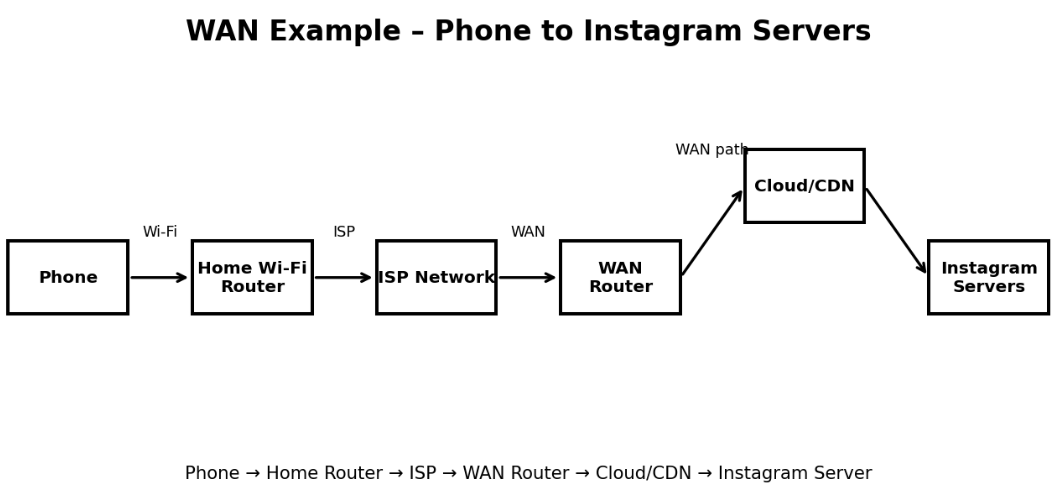
Wide Area Networks

1) Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

WAN Connection Diagram

- The reference diagram shows a typical path from a phone on home Wi-Fi to an Internet application server.
- The home Wi-Fi router connects the local network to the ISP.
- The ISP and Internet/WAN backbone provide the wide-area path toward the destination.
- Routers and cloud/CDN infrastructure can carry traffic toward the application's servers.

Reference Diagram



- Phone → Home Wi-Fi Router → ISP Network → WAN Router → Cloud/CDN → Instagram Servers.

2) Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology Comparison

- MPLS — Speed: High and predictable — Cost: High — Use: Enterprise offices and organizations connecting multiple branches.
- Leased Line — Speed: Dedicated and reliable — Cost: High — Use: Businesses requiring dedicated connectivity between offices or to an ISP.
- Broadband Internet — Speed: Medium to high, depending on service — Cost: Low to medium — Use: Homes, small offices, and general Internet access.

Real-World Examples

- MPLS: A large enterprise with offices in multiple locations may use managed WAN connectivity between branches.
- Leased Line: An organization with critical connectivity requirements may use a dedicated leased connection.
- Broadband Internet: A home user can connect phones, laptops, Smart TVs, and other devices through a broadband ISP.

Comparison Summary

- MPLS is useful when an organization needs managed connectivity across many sites.
- Leased Lines provide dedicated connectivity and predictable performance.
- Broadband is generally the most practical and economical option for homes and small offices.

3) Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Focus on how the VPN creates a private tunnel over the public WAN.

How a VPN Works Over a WAN

- A public Wi-Fi network at a coffee shop provides access to the Internet, which is a public WAN.
- When the user connects to a VPN, the device establishes an encrypted tunnel between the device and a VPN server.
- Traffic from applications such as Spotify or YouTube is sent through this protected tunnel toward the VPN server.
- The VPN server forwards the traffic to the destination Internet service and sends the response back through the VPN connection.
- Encryption helps protect data from local network eavesdropping on an untrusted Wi-Fi network.

Simple Flow

- Phone/Laptop → Public Wi-Fi → Encrypted VPN Tunnel → VPN Server → Internet/WAN → Application Server.

Benefits

- Helps protect traffic from local network eavesdropping on an untrusted Wi-Fi network.
- Provides a protected connection between the user's device and the VPN server.
- Can make Internet services see the VPN server's network as the source of the connection.

4) Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Recommended WAN Technology: MPLS

- MPLS would be a strong choice for a large enterprise connecting warehouses in Ahmedabad, Mumbai, and Delhi.
- Reason 1 — Managed and predictable connectivity: MPLS can provide controlled, reliable connectivity between multiple business locations.
- Reason 2 — Multi-site enterprise networking: MPLS is designed for connecting geographically separated branches and supporting business applications across locations.

Alternative Option

- A modern organization could also consider SD-WAN over multiple broadband or leased connections, depending on cost, availability, security requirements, and application needs.
- For this assignment, MPLS is the recommended answer because the scenario involves multiple geographically separated enterprise sites.

Conclusion

- MPLS is suitable when an organization prioritizes reliable managed connectivity between multiple locations and is willing to pay more than for ordinary broadband.